

Use Case 1 - Main Rewards

Scenario:

1. The player sees a series of **Main Reward Entity** that has spawned in.
2. The player controls the character to walk over the reward.
3. The player's score increases.
4. The **Main Reward Entity** that is “consumed” and is removed from the map.

Exceptions:

1. A **Lethal** or **Non-Lethal Enemy Entity** has entered the same tile as the **Main Reward Entity**. The **Enemy Entity** collision overpowers the ability to “collect” the **Main Reward Entity** and the event is changed to an **Enemy Entity** collision event

Priority: Essential, must be implemented

When Available: On game start

Frequency: Throughout whole game

Channel to Actor: Visual representation

Secondary Actor(s):

Exit Tile

Channel to Secondary Actor(s):

Exit Tile: Requirement Check

Open Issues:

1. **Main Reward Entity** spawn conditions
 - a. Validity checks to ensure all “keys” are available every game.
 - b. Collision checks with other **Rewards** and **Hazard Entities** to avoid multiple events on a specific map tile

Use Case 2 - Additional Rewards

Scenario:

1. The player sees an **Additional Reward Entity** that has spawned in.
2. The player controls the character to walk over the entity.
3. The player's score increases.
4. The **Additional Reward Entity** is “consumed” and is removed from the map.
5. A new **Additional Reward Entity** is created elsewhere on the map.

Exceptions:

1. A **Lethal** or **Non-Lethal Enemy Entity** has entered the same tile as the **Additional Reward Entity**. The **Enemy Entity** collision overpowers the ability to “collect” the **Additional Reward Entity** and the event is changed to an **Enemy Entity** collision event

Priority: Essential, must be implemented

When Available: On game start, with additional periodic generation throughout the game

Frequency: Throughout the whole game

Channel to Actor: Visual representation

Secondary Actor(s):

Other Additional Rewards

Channel to Secondary Actor(s):

Other Additional Rewards: Capacity Update

Open Issues:

1. **Additional Reward Entity** spawn conditions
 - a. Validity checks to ensure any **Additional Reward Entity** is available to be “collected”.
 - b. Collision checks with other **Rewards** and **Hazard Entities** to avoid multiple events on a specific map tile.
 - c. Capacity checks to see if another **Additional Reward Entity** can be created and still align with preset limits.

Use Case 3 - Hazard Penalty

Scenario:

1. The player sees a **Hazard Entity** that has spawned in.
2. The player controls the character to walk over the entity.
3. The player's score decreases.
4. The player's character is “stunned” for a set duration.
5. The **Hazard Entity** is “consumed” and is removed from the map.
6. A new **Hazard Entity** is created elsewhere on the map.

Exceptions:

1. A **Lethal** or **Non-Lethal Enemy Entity** has entered the same tile as the **Hazard Entity**.
The **Enemy Entity** collision overpowers the ability to “collect” the **Hazard Entity** and the event is changed to an **Enemy Entity** collision event

Priority: Essential, must be implemented

When Available: On game start, with additional periodic generation throughout the game

Frequency: Throughout the whole game

Channel to Actor: Visual representation

Secondary Actor(s):

Other Hazards

Channel to Secondary Actor(s):

Other Hazards: Capacity Update

Open Issues:

1. **Hazard Entity** spawn conditions
 - a. Validity checks to ensure any **Hazard Entity** is available to be “collected”.
 - b. Collision checks with other **Rewards** and **Hazard Entities** to avoid multiple events on a specific map tile.
 - c. Capacity checks to see if another **Hazard Entity** can be created and still align with preset limits.

Use Case 4 - Lose Game (Lethal Enemy)

Scenario:

1. The player sees a **Lethal Enemy Entity**.
2. The player's character and the **Lethal Enemy Entity** enter the same tile.
3. The player's character is "caught".
4. The game ends and the current score is totalled and added to the leaderboard.

Exceptions:

1. None, this is an absolute loss condition.

Priority: Essential, must be implemented

When Available: On game start

Frequency: Once per game

Channel to Actor: Visual representation, "Game Over" screen

Secondary Actor(s):

All Game Components:

Rewards,
Hazards,
Character,
etc.

Channel to Secondary Actor(s):

All Game Components: All variables and components are reset to a base value

Open Issues:

1. Reset and Tracking Issues
 - a. All in game components must have proper reset methods to create a smooth user experience.
 - b. Score tracking methods

Use Case 5 - Win Game (Exit)

Scenario:

1. The player collects all **Main Reward Entities**.
2. The player evades all **Lethal Enemy Entities**.
3. The player reaches the marked **Exit Tile** on the map.
4. The game ends and the current score is totalled and added to the leaderboard.

Exceptions:

1. The player enters the **Exit Tile** prior to collecting all **Main Reward Entities**. This will not result in a game ending scenario, the exit tile must act as a normal tile until all **Main Reward Entities** have been collected.

Priority: Essential, must be implemented

When Available: After all **Main Reward Entities** have been collected

Frequency: Once per game

Channel to Actor: Visual representation, "Game Won" screen

Secondary Actor(s):

All Game Components:

Rewards,
Hazards,
Character,
etc.

Channel to Secondary Actor(s):

All Game Components: All variables and components are reset to a base value

Open Issues:

1. Reset and Tracking Issues
 - a. All in game components must have proper reset methods to create a smooth user experience.
 - b. Score tracking methods